Supplementary Material for

Realistic simulation of reflection high-energy electron diffraction patterns for two-dimensional lattices using Ewald construction

Chong Liu,[1,2,a] Kai Chang,[1] and Ke Zou[2,3]

[1]Beijing Academy of Quantum Information Sciences, Beijing 100193, China

[2]Quantum Matter Institute, University of British Columbia, Vancouver, British Columbia V6T 1Z4, Canada

[3]Department of Physics & Astronomy, University of British Columbia, Vancouver, British Columbia V6T 1Z1, Canada

[a]Electronic mail: liuchong@baqis.ac.cn

TABLE S1. Input parameters for the simulations displayed in main text.

<table>
<tr><th></th><th>MgO(001)</th><th>Ge(111)-(2×2)</th><th>STO(001)-($\sqrt{13} \times \sqrt{13}$)</th></tr>
<tr><td>Number of lattices</td><td>1</td><td>2</td><td>2</td></tr>
<tr><td rowspan="2">Lattice base vectors: (x_1,y_1), (x_2,y_2)</td><td rowspan="2">(1,0), (0,1)</td><td>($\sqrt{3}$/2,-1/2), (0,1)</td><td>(2,3), (3,-2)</td></tr>
<tr><td>($\sqrt{3}$,-1), (0,2)</td><td>(3,2), (2,-3)</td></tr>
<tr><td rowspan="2">Structure factors: (r_a,r_b,f)</td><td>(0,0,1)</td><td rowspan="2">(0,0,1)</td><td rowspan="2">(0,0,1)</td></tr>
<tr><td>(0.5,0.5,1)</td></tr>
<tr><td>Lattice unit (Å)</td><td>4.2</td><td>4</td><td>3.9</td></tr>
<tr><td>Grazing angle (°)</td><td>3</td><td>3</td><td>2.6</td></tr>
<tr><td>Rotation (°)</td><td><100>: 0
<110>: 45</td><td><$1\bar{1}0$>: 0
<$1\bar{1}0$>+15°: 15</td><td><100>: 0</td></tr>
<tr><td>Beam E (keV)</td><td>10</td><td>10</td><td>10</td></tr>
<tr><td>Broaden x, y</td><td>1, 1.5</td><td>1, 1</td><td>0.8, 0.7</td></tr>
</table>

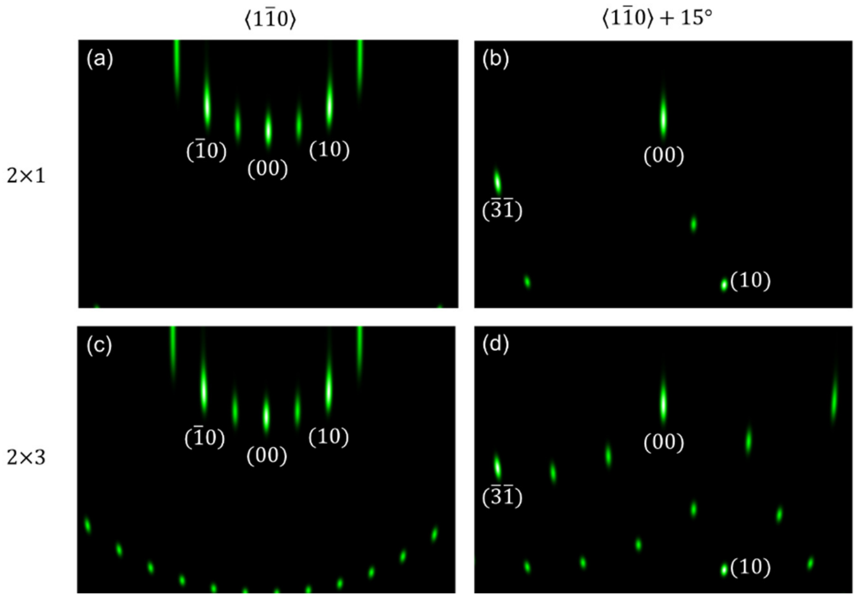

FIG. S1. (a, b) Simulated patterns of 2×1 reconstructed Ge(111) surface with the incident beam along $\langle 1\bar{1}0\rangle$ direction and 15° azimuth from $\langle 1\bar{1}0\rangle$ direction, respectively. (c, d) Simulated patterns of 2×3 reconstructed Ge(111) surface with the incident beam along $\langle 1\bar{1}0\rangle$ direction and 15° azimuth from $\langle 1\bar{1}0\rangle$ direction, respectively.

Running time tests

The code running time increases with increasing number of lattices and reciprocal spots in the plotting range. Taking the most complicated case in this article, $\sqrt{13} \times \sqrt{13}$ reconstruction of STO(001), as an example, the computer system information and testing results are as following.

CPU: Intel(R) Core™ i5-10210U CPU @ 1.60GHz

RAM: 16 GB, 2133 MHz

Operation system: Windows 11

Open up main program interface: 1.0 second

Update reciprocal lattice plot: 5.3 seconds (for preview only)

Generate RHEED simulation and display it in a new window: 6.3 seconds

Update the simulation after changing parameters: 3.0 seconds